subject: **Avaya Communication Manager 6.0**
Call Center DTP
SPI Protocol Enhancements
Element IDs: 10577 10578 10579 10582
Issue: 1.1
COMPAS ID: 139877

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DEVELOPMENT TEST PLAN

1. Introduction

This Development Test Plan describes the procedures for testing SPI Protocol Enhancements provided in the Avaya Communication Manager (CM) 6.0 Call Center release.

The following guidelines were used in formulating test cases included in this plan:

Admin. DEFINITY® Administration Feature DTP Test Case Guidelines, COMPAS ID: 12926[2].

Call Proc. DEFINITY® Call Processing Feature DTP Test Case Guidelines, COMPAS ID: 12929[3].

Unit test cases have not been included in section 9 of this test plan. The following checklist was used in determining the need for and formulating unit test cases included in this plan.

DEFINITY Unit Test Case Checklist, COMPAS ID: 30472[4].

This test plan has been written in accordance with the DTP process defined in:

DTP Process Work Instruction, COMPAS ID: 12925[1].

The DTP process consists of writing and executing the test cases. The execution of the test cases will be recorded using the DEET tool which can be accessed at http://info.dr.avaya.com/defty/deet

1.1 Document Updates.

Version	Date	Description
0.1	2/10/2009	Initial Draft
0.2	4/9/2009	Draft of sections 9.1, 9.2 & 9.3
0.3	4/23/2009	Review copy for sections 9.1, 9.2 & 9.3
0.4	5/11/2009	Incorporate comments from review of sections 9.1, 9.2 & 9.3
0.5	5/15/2009	Section 9.4 and 9.5 added
0.6	5/28/2009	Updated after review
1.0	5/29/2009	Final version for DEET
1.1	4/22/2010	Update for MR 74500 (NOTE: MR 73525 is still open)

2. Document References

2.1 Source Documents

The following documents are those that provide specific input to this document. Each source document has been reviewed.

- Call Center 6.x Product Requirements Document (PRD), CID: 134439 (Section 2.7.5).
- Avaya Communication Manager Call Center 6.0 SRAD/RFS: MIS Link and SPI Protocol Enhancements, Element IDs 10577, 10578, 10579, 10582, CID: 138486.
- Avaya Communication Manager 6.0 Call Center ISD: SPI Protocol Enhancements, Element IDs 10577, 10578, 10579, 10582, CID: 139876.

This set of features is defined by the following element IDs:

10577 Enhance SPI link messages in Language 23 to support Interruptible Aux feature
10578 Enhance SPI link messages in Language 23 to support Route Calls by Skill Level feature
10579 Enhanced reporting support for the Percent Allocation Routing feature
10582 Enhance SPI messages to support increased # of skills per agent

2.2 Related Documents

None.

3. Scope

This test plan covers the CM requirements (10-180) from CID: 138486. The CMS/IQ requirements (CMS10-CMS40) are not covered in this test plan.

4. Terminology

TERM	DEFINITION
Abstract Test #	This is the test case description that is recorded in the DEET tracking tool. This will typically be a unique and concise description of the test. Only text, no graphics or tables.
DEET	DTP Execution Entry Tool. A website application that reads in a DTP or MR_ASSESS, and tracks the state of each Test Case. The website location is **http://info.dr.avaya.com/defty/deet**. Submit the completed DTP to this on-line tool to track test results/status
Requirement Numbers	This can be numbers associated with the Requirement Feature Specifications (RFS), or with the Modification Request (MR).
S8100.	Previously known as Definity One or IP600.
S8300	Previously known as C-hawk
S8700	Previously known as Stingray or Seagull server
Server	Controller or Server of the switch or gateway. For example, G3R, G3SI, G3CSI, S8700,S8300, S8100, ANY or OTHER
Test Case Type BIT	BIT stands for Base Integrity Test. This test type is associated with software base breakage avoidance. These tests are not required under a DTP, but are required in the MR_ASSESS when code is submitted to the base.
Test Case Type Functional	"Idiot" proofing test cases of the new feature ("What if..").
Test Case Type Priority	Basic functionality of the new feature.
Test Case Type Unit	Test cases whose results require special tools

5. Lab System Configuration

Testing will require a CM 6.0 switch connected to either a CMS R16.1 or an IQ 5.1.

6. Special Software/Firmware Needed

No special software or firmware is necessary.

7. Administration Required

No special administration or license files are required.

8. Test Case Format

A test case is the instructions that will be executed to verify one or more requirements. Each test case must contain enough information so the individuals reviewing and executing the plan can determine how to execute the test case and verify the results. For more information on the DTP document, see the DTP process work instructions, COMPAS ID 12925[1].

9. Test Cases

All test cases involving SPI messages can be checked in multiple ways.

- If CMS is getting administration from the switch, the information displayed by CMS can be checked.
- If CMS is changing information on the switch, the new information on the switch can be checked.
- The switch side of the link can be watched using the wlog after executing “test mo l 5126 p 0 t 226” on the SAT.
- The CMS side of the link can be watched using spi.log.

If the tests are run prior to a CMS R16.1 or IQ 5.1 being available, execute the following commands from the SAT:

```
go tcm
rdd :mis_ap
wr Cms_type 19
quit
```

This will force the switch to use language 23. The messages sent from the switch to the CMS can be watched using wlog. The CMS spi.log will not be accurate since it won’t be using language 23. The messages sent from the CMS to the switch cannot be tested with this method.

9.1 SPI Protocol changes for language 23 [RG]

Server: (any)

Test Case Type: Priority

Requirement Numbers: 138486-10

Abstract Test 1: System-parameters Feature-related Form, CMS

Action: On the SAT, go to page 12 of "change system-parameters feature-related". Go to the "CMS (appl mis):" field. Press help.

Verify: The help message includes R16.1.

Action: Enter R16.1. Tab out of the field.

Verify: The value is accepted.

Action: Submit the form.

Verify: The form submits successfully.

Action: Bring the form back up. Go to page 12.

Verify: The CMS field contains R16.1.

Abstract Test 2: System-parameters Feature-related Form, IQ

Action: On the SAT, go to page 12 of "change system-parameters feature-related". Go to the "IQ (appl ccr):" field. Press help.

Verify: The help message includes 5.1.

Action: Enter 5.1. Tab out of the field.

Verify: The value is accepted.

Action: Submit the form.

Verify: The form submits successfully.

Action: Bring the form back up. Go to page 12.

Verify: The IQ field contains 5.1.

Test Case Type: Functional

Abstract Test 3: System-parameters Feature-related Form, CMS & IQ

Action: On the SAT, go to page 12 of "change system-parameters feature-related". Enter R16.1 in CMS and 5.0 in IQ. Submit the form.

Verify: The error message "Allowed CMS/IQ combinations: R16.1/5.1, R15/5.0, R14.1/4.0, R14/4.0 & R13.1/4.0" is displayed.

Action: On the SAT, go to page 12 of "change system-parameters feature-related". Enter R15/R16 in CMS and 5.1 in IQ. Submit the form.

Verify: The error message "Allowed CMS/IQ combinations: R16.1/5.1, R15/5.0, R14.1/4.0, R14/4.0 & R13.1/4.0" is displayed.

Action: On the SAT, go to page 12 of "change system-parameters feature-related". Enter R16.1 in CMS and 5.1 in IQ. Submit the form.

Verify: The form submits successfully.

Test Case Type: Priority

Requirement Numbers: 138486-20

Abstract Test 4: Release and Language in XSTAT20 (CMS)

Action: On the SAT, set CMS release to R16.1 and IQ to blank. During pumpup, check the XSTAT20 message.

Verify: The release specified in the message is 2 (CM6.0).

Verify: The language specified in the message is 0x17 (language 23).

Verify: The dot specified in the message is 0.

Abstract Test 5: Release and Language in XSTAT20 (IQ)

Action: On the SAT, set CMS release to blank and IQ to 5.1. During pumpup, check the XSTAT20 message.

Verify: The release specified in the message is 2 (CM6.0).

Verify: The language specified in the message is 0x17 (language 23).

Verify: The dot specified in the message is 0.

9.2 SPI Protocol changes for Interruptible Aux [RG]

9.2.1 AVAIL & AUDAVAIL

Test Case Type: Functional

Requirement Numbers: 138486-30

Abstract Test 6: Agent goes available without notification, AVAIL23

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. With that agent logged in but not being notified, have the agent go available (either auto-in or manual-in).

Verify: An AVAIL23 message is sent with the Interruptible Aux field set to 0.

Abstract Test 7: AUDAVAIL23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAVAIL23 message is sent with the Interruptible Aux field set to 0.

Test Case Type: Priority

Abstract Test 8: Agent forced auto-in, AVAIL23

Action: On the switch, administer an agent with a skill that has interruptible aux, forced auto-in. With that agent logged in and in AUX, have that skill go into an interrupt state. After the timer expires, the agent is forced to auto-in.

Verify: An AVAIL23 message is sent with the Interruptible Aux field set to 1.

Abstract Test 9: AUDAVAIL23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAVAIL23 message is sent with the Interruptible Aux field set to 1.

Test Case Type: Functional

Abstract Test 10: Agent forced manual-in, AVAIL23

Action: On the switch, administer an agent with a skill that has interruptible aux, forced manual-in. With that agent logged in and in AUX, have that skill go into an interrupt state. After the timer expires, the agent is forced to manual-in.

Verify: An AVAIL23 message is sent with the Interruptible Aux field set to 2.

Abstract Test 11: AUDAVAIL23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAVAIL23 message is sent with the Interruptible Aux field set to 2.

Abstract Test 12: Agent goes available during notification, AVAIL23

Action: On the switch, administer an agent with a skill that has interruptible aux with notification (not forced to available). With that agent logged in and in AUX, have that skill go into an interrupt state. While the agent is being notified, have the agent go available (either auto-in or manual-in).

Verify: An AVAIL23 message is sent with the Interruptible Aux field set to 3.

Abstract Test 13: AUDAVAIL23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAVAIL23 message is sent with the Interruptible Aux field set to 3.

Abstract Test 14: Agent goes available after notify/clear, AVAIL23

Action: On the switch, administer an agent with a skill that has interruptible aux with notification (not forced to available). With that agent logged in and in AUX, have that skill go into an interrupt state. Then have the skill meet the service level target so the notification clears. Then have the agent go available (either auto-in or manual-in).

Verify: An AVAIL23 message is sent with the Interruptible Aux field set to 0.

Abstract Test 15: AUDAVAIL23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAVAIL23 message is sent with the Interruptible Aux field set to 0.

9.2.2 AUX & AUDAUX

Requirement Numbers: 138486-71

Abstract Test 16: Agent goes into AUX, AUX23

Action: On the switch, login an agent. Have the agent go into AUX with a non-interruptible reason code.

Verify: An AUX23 message is sent with the Interruptible Aux field set to 0.

Abstract Test 17: AUDAUX23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAUX23message is sent with the Interruptible Aux field set to 0.

Test Case Type: Priority

Abstract Test 18: Agent goes into AUX, AUX23

Action: On the switch, login an agent. Have the agent go into AUX with an interruptible reason code.

Verify: An AUX23 message is sent with the Interruptible Aux field set to 1.

Abstract Test 19: AUDAUX23

Action: After the previous test case, on the SAT, run "test MO logical 8192 physical 587 test-number 0". This will force an audit.

Verify: An AUDAUX23message is sent with the Interruptible Aux field set to 1.

9.2.3 AEVENT

NOTE: This message is sent by the switch when an agent begins (type=0, button=6) or ends (type=0, button=7) Interruptible Aux notification.

Requirement Numbers: 138486-72 138486-73

Abstract Test 20: (R16.1) skill enters interrupt, AEVENT

Action: (R16.1) Have a skill with more than one agent administered for Interruptible Aux. With the agents in Aux, have the skill go into an interrupt state. Each agent is notified.

Verify: An AEVENT message is sent with type=0, button=6 is sent for each agent that was notified.

Test Case Type: Functional

Abstract Test 21: (R16.1) agent goes into AUX, skill in interrupt, AEVENT

Action: (R16.1) Have a skill in an interrupt state. Have an agent with that skill as interruptible go into AUX. The agent is notified.

Verify: An AEVENT message is sent with type=0, button=6 is sent for the agent that was notified.

Abstract Test 22: (R16.1) agent goes from pending AUX, skill in interrupt, AEVENT

Action: (R16.1) Have a skill in an interrupt state. Have an agent with that skill as interruptible on a call. The agent pushes AUX and goes into a pending AUX state. The agent ends the call and immediately goes into AUX. The agent is notified.

Verify: An AEVENT message is sent with type=0, button=6 is sent for the agent that was notified.

Test Case Type: Priority

Abstract Test 23: (R16.1) skill leaves interrupt, AEVENT

Action: (R16.1) Have a skill with more than one agent administered for Interruptible Aux. With the agents in Aux, have the skill go into an interrupt state. Each agent is notified. Have the skill leave the interrupt state. Each agent stops being notified.

Verify: An AEVENT message is sent with type=0, button=7 is sent for each agent that was cleared.

Test Case Type: Functional

Abstract Test 24: (R16.1) agent changes to AVAIL, AEVENT

Action: (R16.1) Have a skill with an agent administered for Interruptible Aux. With the agent in Aux, have the skill go into an interrupt state. The agent is notified. Have the agent go into AVAIL (either auto-in or manual-in). The agent stops being notified.

Verify: No AEVENT message is sent with type=0, button=7 is sent for the agent.

Abstract Test 25: (R16.1) agent changes to ACW, AEVENT

Action: (R16.1) Have a skill with an agent administered for Interruptible Aux. With the agent in Aux, have the skill go into an interrupt state. The agent is notified. Have the agent go into ACW. The agent stops being notified.

Verify: An AEVENT message is sent with type=0, button=7 is sent for the agent.

Verify: The AEVENT message is sent before the ACW message.

Abstract Test 26: (R16.1) agent changes to AUX, AEVENT

Action: (R16.1) Have a skill with an agent administered for Interruptible Aux. With the agent in Aux, have the skill go into an interrupt state. The agent is notified. Have the agent go into AUX with a non-interruptible reason code. The agent stops being notified.

Verify: An AEVENT message is sent with type=0, button=7 is sent for the agent.

Verify: The AEVENT message is sent before the AUX message.

Abstract Test 27: (R16.1) agent forced to available, AEVENT

Action: (R16.1) Have a skill with more than one agent administered for Interruptible Aux (forced to either auto-in or manual-in). With the agents in Aux, have the skill go into an interrupt state. Each agent is notified. After the timer, each agent stops being notified and is forced to available.

Verify: No AEVENT message is sent with type=0, button=7 is sent for each agent that was cleared.

Abstract Test 28: (R16.1) agent goes into AUX, forced avail, skill in interrupt, AEVENT

Action: (R16.1) Have a skill in an interrupt state. Have an agent with that skill as interruptible (auto-in or manual-in) go into AUX. The agent is notified. After the timer, the agent stops being notified and is forced to available.

Verify: No AEVENT message is sent with type=0, button=7 is sent for the agent.

Abstract Test 29: (R15/R16) skill enters interrupt, no AEVENT

Action: (R15/R16) Have a skill with more than one agent administered for Interruptible Aux. With the agents in Aux, have the skill go into an interrupt state.

Verify: No AEVENT message is sent.

Abstract Test 30: (R15/R16) agent goes into AUX, skill in interrupt, no AEVENT

Action: (R15/R16) Have a skill in an interrupt state. Have an agent with that skill as interruptible go into AUX. The agent is notified.

Verify: No AEVENT message is sent.

Abstract Test 31: (R15/R16) agent goes from pending AUX, skill in interrupt, no AEVENT

Action: (R15/R16) Have a skill in an interrupt state. Have an agent with that skill as interruptible on a call. The agent pushes AUX and goes into a pending AUX state. The agent ends the call and immediately goes into AUX. The agent is notified.

Verify: No AEVENT message is sent.

Abstract Test 32: (R15/R16) skill leaves interrupt, no AEVENT

Action: (R15/R16) Have a skill with more than one agent administered for Interruptible Aux. With the agents in Aux, have the skill go into an interrupt state. Each agent is notified. Have the skill leave the interrupt state. Each agent stops being notified.

Verify: No AEVENT message is sent.

Abstract Test 33: (R15/R16) agent changes work mode, no AEVENT

Action: (R15/R16) Have a skill with an agent administered for Interruptible Aux. With the agent in Aux, have the skill go into an interrupt state. The agent is notified. Have the agent go into a different work mode (AVAIL or ACW). The agent stops being notified.

Verify: No AEVENT message is sent.

Abstract Test 34: (R15/R16) agent forced to available, no AEVENT

Action: (R15/R16) Have a skill with more than one agent administered for Interruptible Aux (forced to either auto-in or manual-in). With the agents in Aux, have the skill go into an interrupt state. Each agent is notified. After the timer, each agent stops being notified and is forced to available.

Verify: No AEVENT message is sent.

Verify: An AVAIL message for each agent.

Abstract Test 35: (R15/R16) agent goes into AUX, forced avail, skill in interrupt, no AEVENT

Action: (R15/R16) Have a skill in an interrupt state. Have an agent with that skill as interruptible (auto-in or manual-in) go into AUX. The agent is notified. After the timer, the agent stops being notified and is forced to available.

Verify: No AEVENT message is sent.

Verify: An AVAIL message for the agent after the AEVENT message.

9.2.4 AGENTSK

NOTE: This message is sent by the CMS/IQ when they want to see an agent's skills (change agent skills, find one).

Requirement Numbers: 138486-40

Abstract Test 36: (R15/R16) AGENTSK20

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, retrieve that agent from the switch.

Verify: The agent is successfully retrieved using AGENTSK20 with the correct data.

Verify: Using SPI logs, the message has the correct structure.

Test Case Type: Priority

Abstract Test 37: (R16.1) AGENTSK23

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R16.1 CMS, retrieve that agent from the switch.

Verify: The agent is successfully retrieved with the correct data.

Verify: Using SPI logs, the new message has the correct structure.

9.2.5 ALOGON

NOTE: This message is sent by the switch when an agent logs in.

Requirement Numbers: 138486-50

Abstract Test 38: ALOGON23

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. Log in the agent.

Verify: A LOGON23 message is sent.

Verify: Using SPI logs, the new message has the correct structure.

9.2.6 SKILLADD

NOTE: This message is sent by the CMS/IQ when they want to change an agent's skills (change agent skills, modify).

Requirement Numbers: 138486-60

Abstract Test 39: (R15/R16) SKILLADD8 – add skill

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, add a skill.

Verify: The skill is successfully added with the correct data on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, add a reserve skill.

Verify: The skill is successfully added with the correct data on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Abstract Test 40: (R15/R16) SKILLADD8 – remove skill

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, remove a non-reserve skill without interruptible aux.

Verify: The skill is successfully removed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, remove a reserve skill.

Verify: The skill is successfully removed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Test Case Type: Functional

Abstract Test 41: (R15/R16) SKILLADD8 – change level

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, Change a skill level a non-reserve skill without interruptible aux.

Verify: The skill level is successfully changed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change a reserve level on a reserve skill.

Verify: The skill level is successfully changed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change a skill with a non-reserve level that uses interruptible aux to a reserve skill.

Verify: The skill level is not changed on the switch.

Verify: The switch reports to CMS that the operation failed at the switch**.**

Abstract Test 42: (R15/R16) SKILLADD8 – to/from reserve

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change a skill without interruptible aux from a non-reserve skill level to a reserve level.

Verify: The skill level is successfully changed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change a skill without interruptible aux from a reserve level on a non-reserve skill level.

Verify: The skill level is successfully changed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Abstract Test 43: (R15/R16) SKILLADD8 – change skill number

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change the skill number of a non-reserve skill without interruptible aux.

Verify: The skill number is successfully changed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change the skill number of a non-reserve skill with interruptible aux.

Verify: The skill number is successfully changed on the switch. The interruptible aux value is cleared out.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R15 CMS, change the skill number of a reserve skill.

Verify: The skill number is successfully changed on the switch.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Abstract Test 44: (R16.1) SKILLADD23

Action: On the switch, administer an agent with multiple skills, some of which have interruptible aux. On R16.1 CMS, change the interruptible aux value for a skill with interruptible aux.

Verify: The agent is successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the change was successful**.**

Verify: Using SPI logs, the new message has the correct structure.

9.2.7 MVAGSTFD

NOTE: This message is sent by the CMS/IQ when they want to change several agent's skills (multi-agent skill change, modify).

Requirement Numbers: 138486-70

Abstract Test 45: MVAGSTFD21 – change skill number

Action: On the switch, administer several agents with multiple skills and skill levels, some of which have interruptible aux. On R15 CMS, change a skill they have in common to a different skill number.

Verify: The agents are successfully changed on the switch with the correct data. If there was an interruptible aux value, it is cleared out.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the message has the correct structure.

Abstract Test 46: MVAGSTFD21 – change skill level

Action: On the switch, administer several agents with multiple skills and skill levels, some of

which have interruptible aux. On R15 CMS, change a skill they have in common to a different skill level.

Verify: The agents are successfully changed on the switch with the correct data. If there was an interruptible aux value, it is cleared out.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the message has the correct structure.

Action: On the switch, administer several agents with multiple skills and skill levels, some of which have interruptible aux. On R15 CMS, change a skill they have in common to a reserve skill level.

Verify: If there was not an interruptible aux value, the agents are successfully changed on the switch with the correct data.

Verify: If there was an interruptible aux value, the switch reports to CMS that the operation failed at the switch for that agent**.**

Verify: Using SPI logs, the message has the correct structure.

Test Case Type: Priority

Abstract Test 47: MVAGSTFD23

Action: On the switch, administer several agents with multiple skills, some of which have interruptible aux. On R16.1 CMS, change the interruptible aux value for a given skill number.

Verify: The agents are successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the new message has the correct structure.

9.3 SPI Protocol changes for Route Calls by Skill Level [RG]

9.3.1 DCON

NOTE: This message is sent by the switch when the call is ringing at the agent.

Requirement Numbers: 138486-80

Abstract Test 48: Skill level in DCON23

Action: With EAS and a measured skill, have a call route to an agent via either a queue or check vector step.

Verify: A DCON23 message is sent with the skill level.

Test Case Type: Functional

Requirement Numbers: 138486-90

Abstract Test 49: Preference match in DCON23, "all-levels"

Action: Have a call route to an agent via a "check…if available-agents…all-lvls" step.

Verify: A DCON23 message is sent with the preference match set to 0.

Abstract Test 50: Preference match in DCON23, "pref-level", match

Action: Have a call route to an agent via a "check…if available-agents… pref-level" step. The agent matches the skill level.

Verify: A DCON23 message is sent with the preference match set to 2.

Abstract Test 51: Preference match in DCON23, "pref-level", no match

Action: Have a call route to an agent via a "check…if available-agents… pref-level" step. The agent does not match the skill level

Verify: A DCON23 message is sent with the preference match set to 1.

Test Case Type: Priority

Abstract Test 52: Preference match in DCON23, "pref-range", match

Action: Have a call route to an agent via a "check…if available-agents… pref-range" step. The agent matches the skill range.

Verify: A DCON23 message is sent with the preference match set to 2.

Test Case Type: Functional

Abstract Test 53: Preference match in DCON23, "pref-range", no match

Action: Have a call route to an agent via a "check…if available-agents… pref-range" step. The agent does not match the skill range.

Verify: A DCON23 message is sent with the preference match set to 1.

Abstract Test 54: Preference match in DCON23, queue-to

Action: Have a call route to an agent via a "queue-to" step.

Verify: A DCON23 message is sent with the preference match set to 0.

Requirement Numbers: 138486-100

Abstract Test 55: non-ACD: Agent surplus flag in DCON23

Action: Have a non-ACD call go to a measured station.

Verify: A DCON23 message is sent with the agent surplus flag set to 0.

Abstract Test 56: ACD: Agent surplus flag "call surplus" in DCON23

Action: Have a call route to an agent after queuing. The agent answers the call.

Verify: A DCON23 message is sent with the agent surplus flag set to 1.

Abstract Test 57: ACD: Agent surplus flag "agent surplus" in DCON23

Action: Have a call route to an agent without queuing. The agent answers the call.

Verify: A DCON23 message is sent with the agent surplus flag set to 2.

9.3.2 VECTOR

NOTE: This message is sent by the switch when the CMS/IQ requests it with a VPARMS to the switch. (Vector administration, find one)

Requirement Numbers: 138486-110

Abstract Test 58: VECTOR23 check step: all levels

Action: On the switch, administer a vector with "check…all-levels". On R16.1 CMS, retrieve that vector from the switch.

Verify: The vector is successfully retrieved with the correct data.

Verify: Using SPI logs, the new message has the correct structure.

Abstract Test 59: VECTOR23 check step: single level

Action: On the switch, administer a vector with "check…pref-level #". On R16.1 CMS, retrieve that vector from the switch.

Verify: The vector is successfully retrieved with the correct data.

Verify: Using SPI logs, the new message has the correct structure.

Test Case Type: Priority

Abstract Test 60: VECTOR23 check step: range

Action: On the switch, administer a vector with "check…pref-range # to #". On R16.1 CMS, retrieve that vector from the switch.

Verify: The vector is successfully retrieved with the correct data.

Verify: Using SPI logs, the new message has the correct structure.

9.3.3 CHGVEC

NOTE: This message is sent by the CMS/IQ to change the vector administration. (Vector administration, modify)

Test Case Type: Functional

Requirement Numbers: 138486-110

Abstract Test 61: CHGVEC23 check step: all-levels w/ < Call Center Release 5.0

Action: On R16.1 CMS, add a vector step of "check…all-levels". Send that change to the switch.

Verify: The vector is successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Abstract Test 62: CHGVEC23 check step: pref-level w/ < Call Center Release 5.0

Action: On R16.1 CMS, add a vector step of "check…pref-level #". Send that change to the switch.

Verify: The vector is not changed on the switch with the correct data.

Verify: The switch reports to CMS that the operation failed at the switch**.**

Abstract Test 63: CHGVEC23 check step: pref-range w/ < Call Center Release 5.0

Action: On R16.1 CMS, add a vector step of "check…pref-range # to #". Send that change to the switch.

Verify: The vector is not changed on the switch with the correct data.

Verify: The switch reports to CMS that the operation failed at the switch**.**

Abstract Test 64: CHGVEC23 check step: all levels

Action: On R16.1 CMS, add a vector step of "check…all-levels". Send that change to the switch.

Verify: The vector is successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Abstract Test 65: CHGVEC23 check step: single level

Action: On R16.1 CMS, add a vector step of "check…pref-level #". Send that change to the switch.

Verify: The vector is successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the new message has the correct structure.

Test Case Type: Priority

Abstract Test 66: CHGVEC23 check step: range

Action: On R16.1 CMS, add a vector step of "check…pref-range # to #". Send that change to the switch.

Verify: The vector is successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the changes were successful**.**

Verify: Using SPI logs, the new message has the correct structure.

9.3.4 VPARMS

NOTE: This message is sent by the CMS/IQ to request a vector. (Vector administration, find one)

Test Case Type: Functional

Requirement Numbers: 138486-120

Abstract Test 67: VECTOR20 check step: all levels

Action: On the switch, administer a vector with "check…all-levels". On R15 CMS, retrieve that vector from the switch.

Verify: The vector is successfully retrieved with the correct data.

Abstract Test 68: VECTOR20 check step: single level

Action: On the switch, administer a vector with "check…pref-level #". On R15 CMS, retrieve that vector from the switch.

Verify: The request is denied with rcode 43 (unsupported step type for this CMS release).

Abstract Test 69: VECTOR20 check step: range

Action: On the switch, administer a vector with "check…pref-range # to #". On R15 CMS, retrieve that vector from the switch.

Verify: The request is denied with rcode 43 (unsupported step type for this CMS release).

9.3.5 CHVEC20

NOTE: This message is sent by the CMS/IQ to change the vector administration. (Vector administration, modify)

Requirement Numbers: 138486-130

Abstract Test 70: CHGVEC20 check step: all levels

Action: On the switch, administer a vector with "check…all-levels". On R15 CMS, change the check step. Send that change to the switch.

Verify: The vector is successfully changed on the switch with the correct data.

Verify: The switch reports to CMS that the changes were successful**.**

Abstract Test 71: CHGVEC20 check step: single level

Action: On the switch, administer a vector with "check…pref-level #". On R15 CMS, change the check step. Send that change to the switch.

Verify: The request is denied with rcode 43 (unsupported step type for this CMS release).

Abstract Test 72: CHGVEC20 check step: range

Action: On the switch, administer a vector with "check…pref-range # to #". On R15 CMS, change the check step. Send that change to the switch.

Verify: The request is denied with rcode 43 (unsupported step type for this CMS release).

9.4 SPI Protocol changes for increased # of skills per agent [GMU]

When testing for messages, make sure as many odd combinations of skills are administered for the agent.
Refer to MR defsw091448.

REF: [j.wallerius] I assume this version of CMS does not support CM6.0, specifically increase of agent skills from 60 to 120. If it does then it might be nice to spread the agent's skill across the skill position range.

9.4.1 AGENTSK

Server: (any)

Test Case Type: Priority

Requirement Numbers: 138486-140

Abstract Test 73: Expand the AGENTSK message into a series message format with up to two messages in the series

Action: CMS generates an APARMS7 via a "find one" command for agent skills to CM so the PBX can respond with a AGENTSK23

Verify: AGENTSK23 message has correct format for various agent skills

Action: execute test for various combinations of Call Center Releases and CMS settings.

Verify: only R16.1 CMS settings will provide AGENTSK23, all other AGENTSKxx will continue to have the correct format and be processed correctly by CMS

Test Case Type: Functional

Abstract Test 74: CCREL 6.0, R15CMS, 120 skills adm on agent

Verify: that only the first 60 skills show up in the AGENTSKxx message

Abstract Test 75: CCREL 6.0, R15 CMS, 60 skills adm on 2nd page of agent form

Verify: that the 60 skills from the second page of agent form show up in the AGENTSKxx message

9.4.2 SKILLADD

Server: (any)

Test Case Type: Priority

Requirement Numbers: 138486-150

Abstract Test 76: Expand the SKILLADD message into a series message format with up to two messages in the series

Action: Use a CMS to change skills for an agent ("modify")

Verify: verify skills change correctly for various settings and values.

Test Case Type: Functional

Abstract Test 77: CCREL 6.0, R15 CMS, 120 skills adm on agent form

Verify: skills change as expected, document the behavior

9.4.3 ALOGON

Server: (any)

Test Case Type: Priority

Requirement Numbers: 138486-160

Abstract Test 78: Expand the ALOGON message into a series message format with up to two messages in the series

Action: login an agent with various skill settings

Verify: verify skill messages in ALOGON is correct

Test Case Type: Functional

Abstract Test 79: CCREL 6.0, R15 CMS, 120 skills adm on agent form

Verify: ALOGON message shows only the first 60 of the 120 skills.

Action: log in an agent with 60 skills on page 2 of the agent form

Verify ALOGON message

Action: log in an agent with 30 skills on page 1 and 30 skills on page 2

Verify ALOGON message

9.4.4 AVAIL, AUDAVAIL, AUDACW, AUDAUX

Server: (any)

Test Case Type: Priority

Requirement Numbers: 138486-170

Abstract Test 80: Double the skill bit maps in the AVAIL, AUDAVAIL, AUDACD and AUDACW messages

Action: Change agent state to AUTO-IN or MANUAL-IN

Verify: verify AVAIL message has correct skill bit mapping

Action: wait for an audit

Verify: the AUDAVAIL message for the correct skill bit mapping

Action: change agent to ACW, wait for audit

Verify: the AUDACW message for the correct skill bit mapping

Action: change agent to AUX, wait for audit

Verify: the AUDAUX message for the correct skill bit mapping

Action: repeat for various settings of skills, CMS Releases and Call Center Releases.

Verify: correct skill bit mappings

Test Case Type: Functional

Abstract Test 81: Try various skill settings measured and unmeasured and repeat the above test case.

9.5 SPI Protocol changes for Percent Allocation Routing [GMU]

Refer to MR defsw090160

9.5.1 DFWD

Server: (any)

Test Case Type: Priority

Requirement Numbers: 138486-180

Abstract Test 82: Add information about a call being redirected by a VDN with a PRT table to the DFWD SPI message

Action: Make a call to a measured VDN with PRT

Verify: DFWD SPI message matches requirements

Test Case Type: Functional

Abstract Test 83: Try other call type scenarios with different aspects of the call measured or not.

- Originate on
  - measured/unmeasured Trunk
  - measured agent, unmeasured station
- Call a VDN with PRT that is measured/unmeasured
  - Consider enabling/disabling VDN Override
  - Note the DFWD to the VDN with PRT will have vec_num=0
- Routing point has a
  - measured/unmeasured VDN/vector
    - again consider VDN Override settings
  - measured/unmeasured VDN/PRT
    - again consider VDN Override settings
  - Verify the DFWD at this point will have PRT number, and VDN routing internal line number, and the Routing point Target Percentage set.

Abstract Test 84: Consider testing with covering to a VDN with PRT

Abstract Test 85: Test toggling all fields in the DFWD message [e.smelko]

10. Notes

References

1. MultiVantage™ DTP Process Work Instruction , COMPAS ID: 12925
2. DEFINITY® Administration Feature DTP Test Case Guidelines, COMPAS ID: 12926.
3. DEFINITY® Call Processing Feature DTP Test Case Guidelines, COMPAS ID: 12929.
4. DEFINITY® Unit Test Case Checklist, COMPAS ID: 30472.

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